

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks: 25

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I P. Harsha Vardhan ¹⁹²³¹¹⁰³⁹ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

P. Harsha Vardhan
Signature of the student:

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C? (5 Marks)
2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C? (5 Marks)
3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis. (5 Marks)
4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)
 - (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
 - (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
 - (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
 - (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
 - (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
 - (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
 - (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
 - (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

- i) Identify the epoch at which the model begins to overfit.
- ii) Suggest two effective techniques to reduce overfitting in this model.
- iii) Discuss how overfitting affects the model's generalization ability on unseen data.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

- (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
- (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
- (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
- (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
- (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
- (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
- (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
- (h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%
- i) Identify the epoch at which the model starts overfitting.
- ii) Explain why validation accuracy decreases despite training accuracy increasing.
- iii) Suggest two methods to improve validation performance.

6. A machine learning model shows the following loss values during training: (5 Marks)

- (a) Epoch 1: Training Loss = 3.0, Validation Loss = 2.7
- (b) Epoch 2: Training Loss = 2.6, Validation Loss = 2.3
- (c) Epoch 3: Training Loss = 2.3, Validation Loss = 2.1
- (d) Epoch 4: Training Loss = 2.1, Validation Loss = 2.0
- (e) Epoch 5: Training Loss = 1.9, Validation Loss = 2.2
- (f) Epoch 6: Training Loss = 1.7, Validation Loss = 2.5
- (g) Epoch 7: Training Loss = 1.6, Validation Loss = 2.8
- (h) Epoch 8: Training Loss = 1.4, Validation Loss = 3.1
- i) At which epoch does the model achieve optimal generalization?
- ii) Identify signs of overfitting from the data.
- iii) Recommend two regularization techniques and explain their impact.

7. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires). (5 Marks)

- a) Front identified as Front: 119
- b) Front identified as Back: 06
- c) Back identified as Back: 120
- d) Back identified as Front: 05

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

8. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks)

- a) Diseased identified as Diseased: 138
- b) Diseased identified as Healthy: 12
- c) Healthy identified as Healthy: 135
- d) Healthy identified as Diseased: 15

iii.
iv.

Compute: Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.

9. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam): (5 Marks)

- Spam identified as Spam: 180
- Spam identified as Not Spam: 20
- Not Spam identified as Not Spam: 170
- Not Spam identified as Spam: 30

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

10. In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non-defective): (5 Marks)

- Defective identified as Defective: 230
- Defective identified as Non-Defective: 20
- Non-Defective identified as Non-Defective: 225
- Non-Defective identified as Defective: 25

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

11. In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated. (5 Marks)

12. A warehouse has two sections. Section A contains 15 defective items and 35 non-defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective. (5 Marks)

13. In a binary classification problem using a deep learning model with a K-Nearest Neighbor classifier, the model produces True Positive (TP) = 120 and True Negative (TN) = 150. The dataset contains 1000 samples per class, and 20% of the data is used for testing. (5 Marks)
Find the following: Accuracy, Precision, Sensitivity (Recall), Specificity.

14. A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data. Calculate: Accuracy Precision Sensitivity Specificity. (5 Marks)

15. A deep learning model is trained for a classification task, and the following loss values are observed: (5 Marks)

- Epoch 1: Training Loss = 2.8, Validation Loss = 2.6
- Epoch 2: Training Loss = 2.4, Validation Loss = 2.2

- iii. Epoch 3: Training Loss = 2.1, Validation Loss = 1.9
- iv. Epoch 4: Training Loss = 1.9, Validation Loss = 1.8
- v. Epoch 5: Training Loss = 1.7, Validation Loss = 1.9
- vi. Epoch 6: Training Loss = 1.5, Validation Loss = 2.1
- vii. Epoch 7: Training Loss = 1.3, Validation Loss = 2.4
- viii. Epoch 8: Training Loss = 1.2, Validation Loss = 2.7

Answer the following:

- a. At which epoch does the model achieve the best generalization performance?
- b. At which point does overfitting begin? Justify your answer.
- c. Suggest two techniques to reduce overfitting and briefly explain how they help.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue and 2 green balls. Box B contains 5 red, 7 blue and 3 green balls. Box C contains 8 red, 2 blue and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes from the chosen box is found to be blue. What is the probability that the selected box was Box.

Problem context:

Box A: 6R 4B 2G (12 balls)

Box B: 5R 7B 3G (15 balls)

Box C: 8R 2B 5G (15 balls)

B is twice as likely as A or C, so weights are
1 : 2 : 1

$$P(A) = \frac{1}{4}$$

$$P(B) = \frac{1}{2}$$

$$P(C) = \frac{1}{4}$$

→ Evidence: The ball drawn is blue

* Find: $P(C|Blue)$

Concept Applied:

Bayes theorem: (reverse reasoning)

$$P(C|Blue) = \frac{P(Blue|C) \cdot P(C)}{P(Blue)}$$

Total probability

$$P(\text{Blue}) = \sum P(\text{Blue}|X) \cdot P(X), X = A, B, C$$

Likelihoods (fraction of blue)

$$P(\text{Blue}|A) = \frac{4}{12} = \frac{1}{3}$$

$$P(\text{Blue}|B) = \frac{7}{15}$$

$$P(\text{Blue}|C) = \frac{2}{15}$$

Problem-solving

Joint probability:

$$A: \frac{1}{4} \times \frac{1}{3} = \frac{1}{12} = \frac{5}{60}$$

$$B: \frac{1}{2} \times \frac{7}{15} = \frac{7}{30} = \frac{14}{60}$$

$$C: \frac{1}{4} \times \frac{2}{15} = \frac{1}{30} = \frac{2}{60}$$

$$P(\text{Blue}) = \frac{5+14+2}{60} = \frac{21}{60} = \frac{7}{20}$$

$$\text{Posterior of } C = \text{joint}(C) / P(\text{Blue}) = \frac{\frac{2}{60}}{\frac{21}{60}} = \frac{2}{21}$$

$$P(C|\text{Blue}) = \frac{2}{21} \approx 0.095 = 9.5\%$$

Interpretation: C is the least likely source. It has a

low prior ($\frac{1}{4}$) & the lowest blue fraction ($\frac{2}{15}$).

Box B is the most likely origin ($\frac{14}{21} \approx 66.7\%$) because it is favoured twice over and holds many blue balls

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue and 2 black cars; Section B has 4 red, 6 blue and 5 black cars; Section C has 7 red, 2 blue & 6 black cars. A section is chosen at random, but section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen sections turns out to be blue. What is the probability that car came from the section C?

Problem context:

Section A : 5R 3B 2k (10 cars)

B = blue

Section B : 4R 6B 5k (15 cars)

k = Black

Section C : 7R 2B 6k (15 cars)

R = red

B is chosen twice as often : weight 1 : 2 : 1

$$P(A) = \frac{1}{4} \quad P(B) = \frac{1}{2} \quad P(C) = \frac{1}{4}$$

Evidence: the car is blue Find: $P(C|B\text{blue})$

Concept applied:

Bayer's theorem with total probability

$$P(C|B\text{blue}) = \frac{P(B\text{blue}|C) P(C)}{\sum P(B\text{blue}|X) \cdot P(X)}$$

Likelihoods (fractions of blue cars)

$$P(\text{Blue}|A) = \frac{3}{10} \quad P(\text{Blue}|B) = \frac{6}{15} = \frac{2}{5} \quad P(\text{Blue}|C) = \frac{2}{15}$$

Prior $\times$ likelihood = each section's
shared of all blue cars.

Problem-solving:

$$\text{Joint probabilities: } A: \frac{1}{4} \times \frac{3}{10} = \frac{3}{40} = \frac{9}{120}$$

$$B: \frac{1}{2} \times \frac{2}{5} = \frac{1}{5} = \frac{24}{120}$$

$$C: \frac{1}{4} \times \frac{2}{15} = \frac{1}{30} = \frac{4}{120}$$

$$P(\text{Blue}) = \frac{9+24+4}{120} = \frac{37}{120}$$

$$P(C|\text{Blue}) = \frac{4}{120} \times \frac{120}{37} = \frac{4}{37}$$

$$P(C|\text{Blue}) = \frac{4}{37} \approx 0.108 \approx 10.8\%$$

Interpretation: Section C is unlikely to be the source because it has few blue cars (2 of 15) & low prior. Section B dominates $24/37 = 64.9\%$ due to its double usage and 6 blue cars.

3 In a real-world medical image classification task for detection of breast cancer, a deep-learning model integrated with the k-Nearest Neighbours algorithm is applied for the binary classification (malignant vs benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing during evaluation, the model reports 210 true positives (TP), 190 True Negatives (TN) and 30 False positives (FP). Based on the given information, determine the missing false negatives (FN) values, and then calculate the accuracy, precision, Recall & Specificity of classifier, explaining its effectiveness in medical diagnosis.

KNN + Deep Learning for Breast Cancer

Problem context:

Given, 1800 samples per class $\rightarrow$ 3600 total.

20% test set = 720 samples.

TP = 210 TN = 190 FP = 30 FN = ?

Task: find FN, then accuracy, precision, sensitivity, specificity and judge usefulness of diagnosis (Positive = malignant).

Concept Applied:

Confusion-matrix identifies:

$$TP + TN + FP + FN = \text{test size}$$

$$\text{Accuracy} = TP + TN / \text{Total}$$

$$\text{Precision} = TP / TP + FP$$

$$\text{Sensitivity} = TP / TP + FN$$

$$\text{Specificity} = TN / TN + FP$$

$$FN = 720 - (40 + 190 + 30) = 290$$

Confusion matrix:

Pred+
Actual+ TP = 210

Pred-
FN = 290

Actual- FP = 30

TN = 190

$$\text{Actual+} = 210 + 290 + 500$$

$$\text{Total} = 720$$

$$\text{Actual-} = 30 + 190 = 220$$

$$\begin{array}{l} \text{Accuracy} \\ \frac{210 + 190}{720} \\ = 55.6\% \end{array}$$

$$\begin{array}{l} \text{Precision} \\ \frac{210}{210 + 30} \\ = 87.5\% \end{array}$$

$$\begin{array}{l} \text{Sensitivity} \\ \frac{210}{210 + 290} \\ = 42.0\% \end{array}$$

$$\begin{array}{l} \text{Specificity} \\ \frac{190}{190 + 30} \\ \approx 86.4\% \end{array}$$

Effectiveness is medical diagnosis: Precision (87.5%) & specificity (86.4%) are high. so the model raises few false alarms are rarely sends healthy patients for unnecessary biopsy. However, sensitivity is only 42%: 290 of 500 malignant cases were missing. In Cancer screening a missed tumor is far costlier than a false alarm. so this classifier is not yet safe for stand-alone diagnosis. The 55.6% accuracy is also modest and hides this imbalance in errors.

4. Examine the performance of a ML model using its training & validation loss values over multiple epochs.

- a) E1: TL = 2.8 VL = 2.4
- b) E2: TL = 2.5 VL = 2.2
- c) E3: TL = 2.2 VL = 2.0
- d) E4: TL = 2.0 VL = 2.1
- e) E5: TL = 1.8 VL = 2.3
- f) E6: TL = 1.6 VL = 2.6
- g) E7: TL = 1.5 VL = 2.9
- h) E8: TL = 1.3 VL = 3.2

E = Epoch
TL = Training Loss
VL = Validation Loss

- 1) Suggest two effective techniques to reduce overfitting in this model.
- 2) Identify the epoch at which the model begins to overfit.
- 3) Discuss how overfitting affects the model's generalization ability on unseen data.

Problem context & concept:

Loss measures error (Lower = better). Healthy learning: both curves fall together. Overfitting: training loss keeps falling while validation loss turns towards i.e., the model memorises training data instead of learning general patterns. Look for the minimum validation loss and the point where the gap starts to widen.

2.(i) Epoch where overfitting begins

Validation loss falls until Epoch 3 (2.0, the minimum) then rises at Epoch 4 (2.1) while training loss still drops (2.2 $\rightarrow$ 2.0).

So the model begins to overfit at Epoch 4. Epoch 3 is the best checkpoint. The gap grows from 0.2 (Ep 3) to 1.9 (Ep 8: 3.2 vs 1.3)

3 (ii) Two techniques to reduce overfitting

(1) Early stopping: Monitor validation loss and stop at Epoch 3 (restore the best weights)

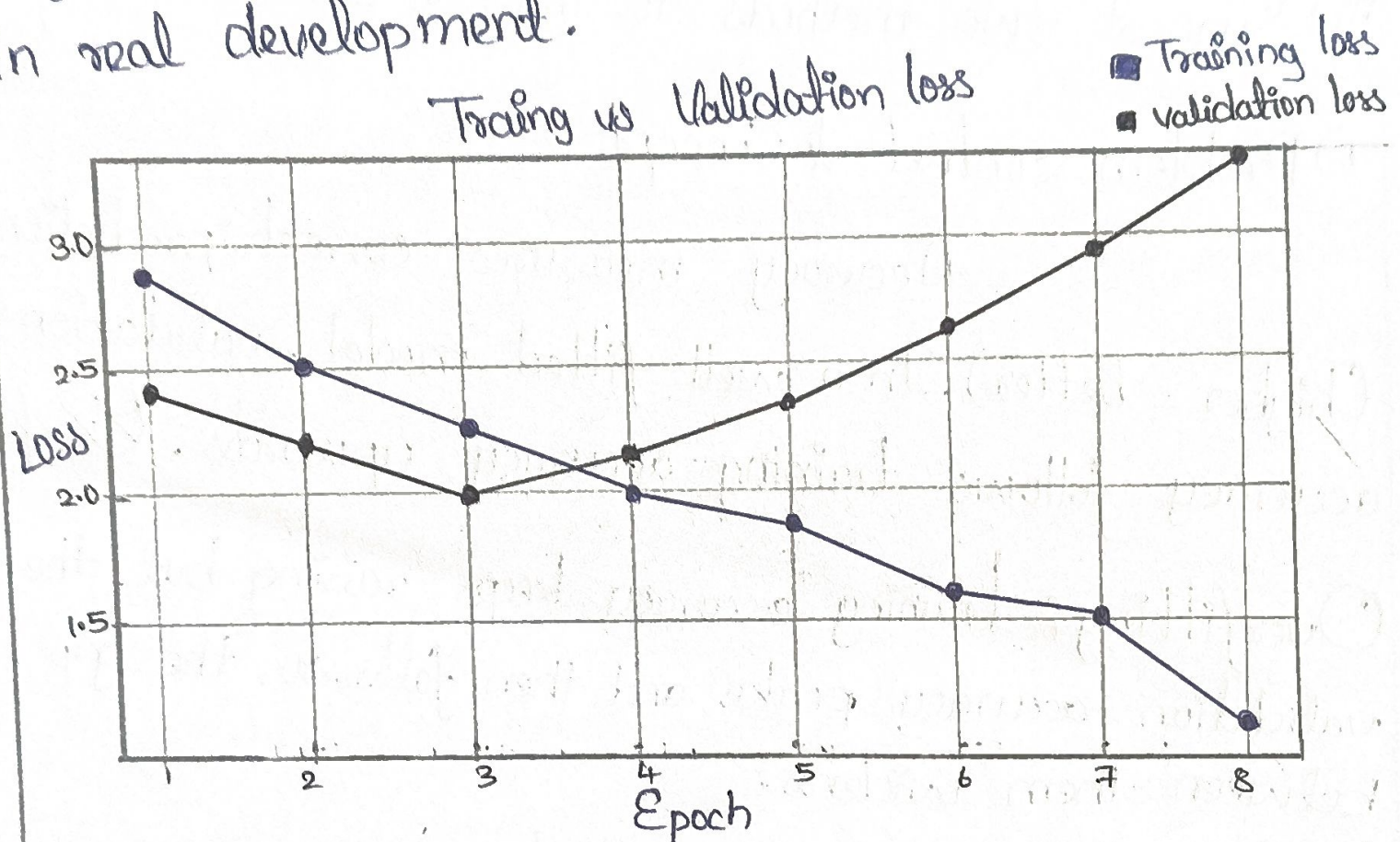
(2) Regularisation (Dropout / L2 weight decay): randomly drops neurons (or) penalises large weights so the network cannot memorise noise.

helps: Data augmentation, more training data, Small model

4 (iii) Effect on generalization.

An overfitting model has very often low training error but high error on unseen data. It learns noise and quirks of the training set (high variance) so predictions on new inputs become unreliable.

the widening train-validation gap is the generalised gap. Its good training scores gives a false sense of quality, so it will perform poorly in real development.



5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs.

- a) E1: TA = 60% VA = 58%
- b) E2: TA = 65% VA = 63%
- c) E3: TA = 70% VA = 68%
- d) E4: TA = 75% VA = 72%
- e) E5: TA = 80% VA = 73%
- f) E6: TA = 85% VA = 72%
- g) E7: TA = 88% VA = 70%
- h) E8: TA = 92% VA = 68%
- B)

- i) Identify the epoch at which the model starts overfitting
- ii) Explain why validation accuracy decreases despite training accuracy increasing.
- iii) Suggest two methods to improve validation performance

1) Problem context & Concept:

Accuracy measures correct predictions (higher = better). In a well-fitted model, validation accuracy follows training accuracy upwards. ~~Overf~~

Overfitting: training accuracy keeps raising but the validation accuracy peaks and then falls, so, the gap between them widens.

2. Epoch where overfitting starts

Validation accuracy rises to its peak of 73% at Epoch 5, where the gap accuracy jumps from 3 to 7 points; it then falls from Epoch 6 (72%) while training accuracy keeps climbing. So the model starts overfitting after Epoch 5 (Epoch 5 is the best model; the decline is clear from the Epoch).

3) Why validation accuracy drops

The model keeps fitting details and noise specific to the training samples (memorisations) rather than patterns that hold for new data. Training accuracy only reflects seen data so it keeps improving, while validation accuracy measures unseen data and drops as the model becomes over-specialized. The gap grows from 2 points (Ep 1) to 24 points (Ep 8: 92% vs 68%)

4) Two methods to improve validation performance

1. Early stopping at Epoch 5 (keep the best validation point)
2. Dropout / L2 regularisation to limit model complexity

Also: data augmentation, more training data, simpler architecture or k-fold cross validation for the reliable turning.

Training vs validation accuracy

